

SOUTH KOREA TECHNOLOGY: SEMICONDUCTORS - MEMORY

Addressing 8 key questions on memory including HBM pricing, LTA, inventory, China DRAM, and ADR; reiterate Buy on Samsung (on CL) and SK Hynix

The share prices of Samsung Electronics (SEC) and SK Hynix have declined by 23%/35% during the past month. While acknowledging there are several factors outside of the industry/company fundamentals that may have contributed to this, from the fundamentals perspective we believe the correction can be attributed to the concerns including: 1) weakening memory pricing expectations, 2) lack of LTA details, 3) higher inventory at module makers, 4) increasing China supply, 5) lack of shareholder return announcements, and 6) Hynix's 2Q26 OP miss.

Post the correction, SEC is trading at 3.6x 2027E P/E and 1.4x P/B, with Hynix at 3.5x P/E and 1.6x P/B, which implies the market does not believe in the sustainability of the solid earnings for these firms. In this report we address 8 key investor questions on the memory industry, and reiterate our Buy ratings on SEC (on CL) and Hynix.

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Q1. How much upside can we expect for 2027 HBM pricing? -> We expect a ~100% yoy increase.

Q2. How are the LTAs being structured, and how are they favorable for the suppliers? -> We believe LTAs are seeing longer duration and larger coverage.

Q3. Are the recent concerns on elevated inventory valid? -> We don't think so given customer/supplier inventory levels are healthy.

Q4. Is NAND turning into oversupply? -> We don't think so, and expect server SSD demand to offset any weakness in consumer.

Q5. How will CXMT impact the ongoing memory supply shortage? -> CXMT to increase its presence mainly for local China demand, while export usage driven by both commercial and geopolitical considerations.

Q6. How will Korean players respond to increasing shareholder return demand? -> We expect upside in the Street's shareholder return expectations

Q7. How will Hynix's U.S. ADR impact its domestic shares? -> We expect the ADRs will likely continue to trade at a premium over the domestic shares, but Hynix shares to gradually resolve the historical discount that it has received relative to peers.

Q8. Why did Hynix miss 2Q26 and is this trend going to continue in 2H26? -> Due to lower DRAM ASP, but we expect solid growth in both volume and ASP in 3Q26.

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Addressing 8 key investor questions on memory

Q1. How much upside can we expect for 2027 HBM pricing?

We expect HBM blended ASP for SEC/Hynix to reach close to US\$ 2.9/Gb in 2027E, which would be roughly an 87%/100% yoy increase, respectively. We expect the like-for-like basis price increase to be roughly 60% for major products, and the remaining growth to come from the mix improvement impact.

We expect the strong HBM demand led by AI servers to continue to outpace supply, and highlight the rising difficulty in ramping the latest generation HBM. This comes from the rising complexity in using a more advanced node for both core die and base die, as well as moving to a higher layer HBM stack, which results in a lower production yield. The higher trade ratio between HBM and conventional DRAM also leads to incrementally less supply of HBM as we move forward to the new generation of HBM. As such, we expect 2027 S/D to be even tighter compared to this year, which should help drive a favorable pricing environment.

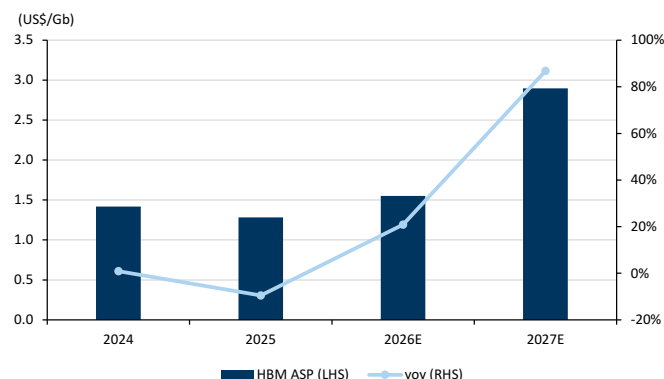
Another reason why we expect a significant HBM pricing increase next year is the material gap between HBM and conventional DRAM. While HBM is more of a “premium” DRAM used for AI servers and conventional DRAM is a “regular” DRAM used for various applications including smartphones/PCs/servers, given the different contract dynamics where the former is typically negotiated on an annual basis and hence mostly fixed throughout the year but the latter on a monthly/quarterly basis and hence can bake in market dynamics more frequently, as of 2Q26 the pricing for conventional DRAM has exceeded that of HBM.

By the end of this year, we expect average conventional DRAM pricing to rise to around US\$2/Gb from US\$0.5–0.6/Gb at the end of 2025, and hence expect a significant HBM pricing increase for next year so that it can sell at a “premium” and also reach a closer operating margin level considering the higher cost structure vs. conventional DRAM. The Visible Alpha consensus estimate for Hynix HBM ASP is around US\$2.3/Gb, and hence we are 24% higher than the Street.

With the significant pricing increase we expect, we estimate HBM revenue mix as a % of SEC/Hynix DRAM revenue will rise to 16%/22% in 2027E and 18%/25% in 2028E from 8%/14% this year, and expect the more stable demand/pricing and improved visibility coming from the product to help drive a higher valuation multiple for the memory names.

Exhibit 1: We expect HBM blended ASP for SEC to reach close to US\$ 2.9/Gb in 2027E, implying an 87% yoy increase

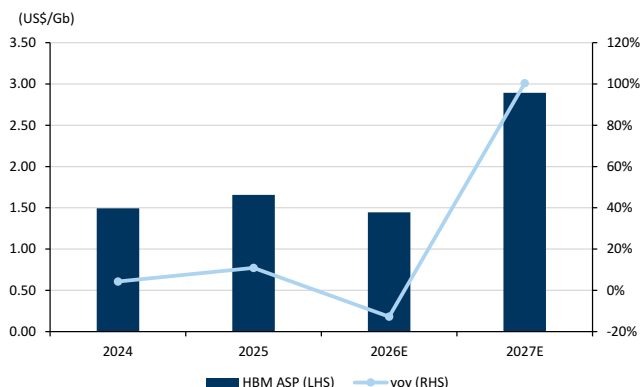
SEC HBM ASP and yoy growth



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 2: We expect HBM blended ASP for Hynix to reach close to US\$ 2.9/Gb in 2027E, implying a 100% yoy increase

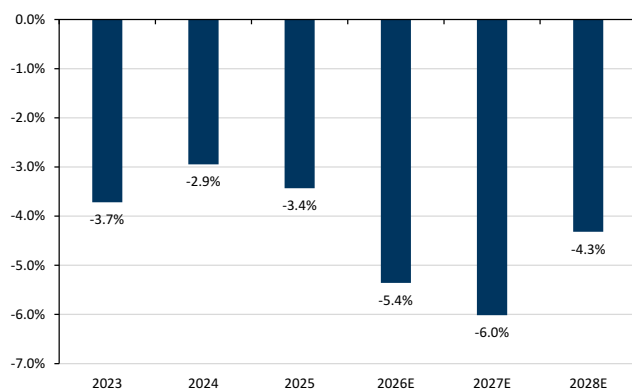
Hynix HBM ASP and yoy growth



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 3: We expect 2027 HBM S/D to be even tighter compared to this year, and help drive a favorable pricing environment.

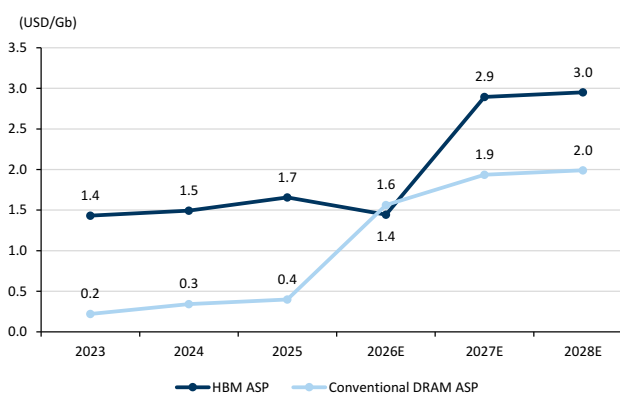
HBM industry S/D



Source: Goldman Sachs Global Investment Research

Exhibit 4: We expect a significant HBM pricing increase for next year

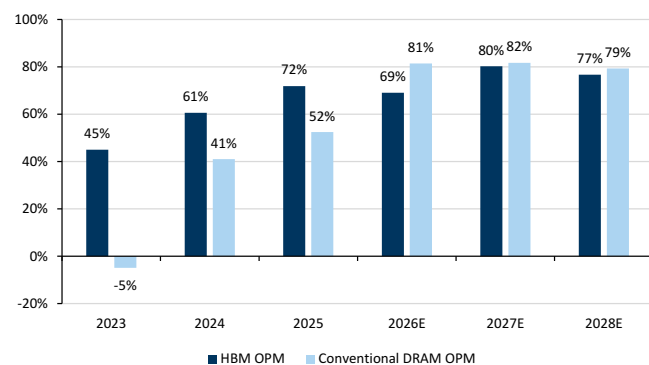
Hynix conventional DRAM ASP vs. HBM ASP



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: We expect HBM to reach a closer operating margin level to conventional DRAM considering the higher cost structure

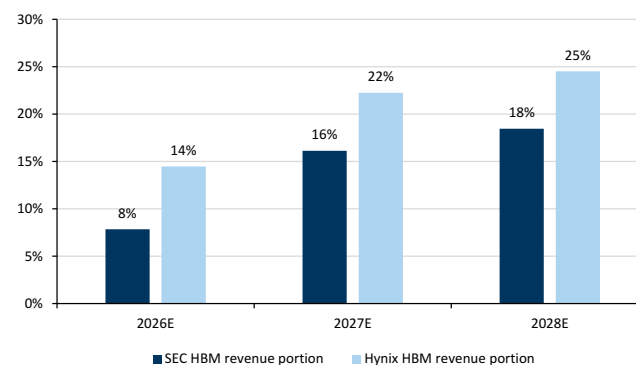
Hynix conventional DRAM OPM vs. HBM OPM



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 6: We estimate HBM revenue mix as a % of SEC/Hynix DRAM revenue will rise to 16%/22% in 2027E and 18%/25% in 2028E from 8%/14% this year

SEC/Hynix HBM as % of DRAM revenue



Source: Company data, Goldman Sachs Global Investment Research

Q2. How are the LTAs being structured, and how are they favorable for the suppliers?

As expectations for a prolonged and substantial memory supply tightness have continued to increase, customers have been actively asking for LTAs (Long-Term Agreement; also called New Business Model or Strategic Customer Agreements) to secure mid-to-long term volume, while suppliers are also engaging in the discussions as LTAs can enhance their long-term pricing/earnings visibility and business stability.

While acknowledging there are still plenty of details that have not been disclosed and the terms of these LTAs would be different depending on the match between the customer and supplier, based on what has been disclosed and our checks, we believe as time goes by the terms of the LTAs are becoming more favorable for the suppliers through: **1) longer duration, 2) larger coverage, 3) pricing schemes that reduce volatility, and 4) stronger binding power.**

Duration: 3–5 years, potentially longer with rolling contracts

Most suppliers have indicated that the LTAs are mostly signed on a 5-year basis, with 3 years for certain customers. **SEC** also mentioned at its earnings call last week that while its LTA discussions are typically based on a 5-year term, it is structured as rolling contracts where negotiations would be conducted each year to extend the contract period by an additional year. So while we believe the initial expectations of the duration of LTAs have been 3–5 years, there is option to extend even beyond 5 years as these rolling mechanism could potentially lengthen the contract duration.

LTA coverage: target increasing to 60–70%, which offers higher long-term visibility

At its earnings call in early-May, **SanDisk** mentioned it has signed NBM contracts with 5 customers which cover 1/3 of its bits for CY27, and that it targets to cover 50% of its volume in the long term through NBMs. **Micron** at its earnings call back in late-June mentioned that it has signed 16 SCAs, which account for 20% of its DRAM volume and 1/3 of its NAND volume over the contract term, and expects SCAs to eventually account for 50%+ of its revenue. At its earnings call last week, **Hynix** mentioned that it concluded around 10 LTA negotiations including key customers. Last week **SEC** disclosed that it has completed contracts with the top five global data center customers, and is currently in the final stage of negotiations with five additional large customers related to AI demand. Once the contracts are finalized, multi-year contract volumes are expected to reach approximately 60–70% of its planned production capacity.

While the companies that initially disclosed some details of the LTAs have set the target of LTA coverage at around 50% of volume/revenue, the most recent LTA coverage target (by SEC) has increased to 60–70% which we believe is positive given that it offers improved long-term demand visibility and business stability. In addition, despite recent large long-term capex announcements leading to some concern of oversupply, the company mentioning LTAs covering 60–70% of its future planned production capacity offers increased assurance that the significant spending would be to meet the significant demand growth that are visible through these LTAs, and not easily lead to oversupply.

Pricing: moving towards more favorable structures such as pricing bands and floor pricing

We believe there are three main types of pricing schemes that are being discussed,

which are: 1) fixed pricing, 2) pricing bands (with floor and ceiling), and 3) floor pricing, where floor pricing would be the most favorable for the suppliers with fixed being less favorable. Under pricing bands, an abrupt pricing increase or decrease would be difficult, hence would likely reduce the pricing volatility that exists without a LTA. With a floor pricing, the downside for the suppliers is limited while they would be able to capture any upside in market pricing.

At its earnings call back in early-May, **SanDisk** mentioned that longer-term contracts have more variable pricing while shorter-term contracts have more fixed pricing. **Micron** at its earnings call back in late-June mentioned that its largest agreements would have pricing bands with floor and ceiling price (set at current 2QCY26 market price), where gross margin for floor price would still be higher than previous peak margin. The company mentioned other agreements would be based on fixed pricing. **Hynix** last week mentioned that pricing structures are designed to address price volatility, where we believe the company is discussing all three pricing schemes mentioned above.

SEC also mentioned at its earnings call last week that the company aims to reflect a pricing structure that can offset future investment risk to a certain extent, where different pricing models are applied depending on the customer group and product category. For general-purpose products, the company has set floor prices at a level sufficient to prepare for future investment risks that may arise from fluctuations in market prices. We believe this likely implies that the company is mainly going after more favorable pricing structures in either pricing bands or floor pricing.

Binding commitments: meaningful prepayments/deposits leading to stronger binding power

We believe one of the key differences of the LTAs during this cycle compared to previous ones is that they include sizable prepayments and meaningful penalty clauses for non-compliance which guarantee stronger binding power. The main methods of commitment that make the contracts more binding are advanced payment method and take or pay method.

Highlighting comments made at the most recent earnings call by each memory supplier, **SanDisk** mentioned that financial guarantee exceeds US\$11bn including prepayments. **Micron** mentioned take-or-pay agreements with binding commitments to purchase specific volumes are included in contracts, and the company expects to receive US\$22bn in cash deposit and related financial commitments, where the deposits will be returned to customers over time, primarily in the latter half of the agreement. **Hynix** mentioned that financial mechanisms (e.g. deposits) are incorporated to support contract fulfillment, enhancing mid-to-long-term demand visibility. **SEC** commented that to strengthen the binding nature of contracts, the company is including substantial advance payments in the form of deposits held over multiple years, and it has already received roughly 1/4 of total contractual advance payments. If additional contracts proceed, the size of advance payments is also expected to increase further, but the specific amount of advance payments is difficult to disclose.

Exhibit 7: Summary of LTA Terms

Term	Samsung Electronics	SK Hynix	Micron	SanDisk
Duration	5 years, with rolling annual extension	Mostly 5 years	Mostly 5 years, 3 years for automotive clients	-
Coverage <i>signed to date</i>	Top five data-center customers signed; five more in final talks	10+ LTAs concluded, key customers included	16 SCAs: 20% of DRAM and 1/3 of NAND volume	5 customers, 1/3 of CY27 bits
Coverage <i>target</i>	60–70% of planned production capacity	-	50%+ of revenue	50% of volume
Pricing	Floor prices; varies by customer	-	Bands with floor and ceiling (CQ2 market level)	Longer contracts more variable-priced
Binding power	Multi-year deposits; ~1/4 already received	-	Take-or-pay; ~US\$22bn of deposits	\$11bn+ of guarantees with prepayments

Direction of travel — every term has moved in the suppliers' favour				Initial terms	Where they are heading
Duration	3–5 years	➡	5-year base with rolling annual extension, with potential to run beyond 5 years		
Coverage	~50% of volume targeted	➡	60–70% of volume, measured against planned future capacity		
Pricing	Mostly fixed pricing	➡	Pricing bands with floor and ceiling with favorable margins		
Binding power	Limited commitment	➡	Meaningful advance payments and take-or-pay obligations		

Source: Company data, Goldman Sachs Global Investment Research

Q3. Are the recent concerns on elevated inventory valid?

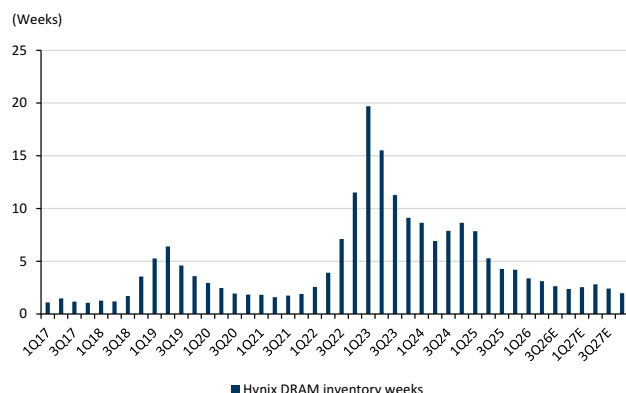
Recently there have been rising concerns of elevated inventory at the module makers. Based on our checks, we do not disagree that inventory level at the module makers have been rising, especially with the relatively weak demand from consumer products such as smartphones/PCs. At the same time, we believe this likely has a larger impact on sentiment rather than the industry fundamentals, given this market comprises only a single-digit% portion of the entire memory market. In addition, we believe the inventory level of the major memory suppliers and customers are at a healthy level.

For the suppliers, as of end-2Q26 we estimate inventory levels for both DRAM and NAND are within 2–4 weeks, below the normal level of around 4–5 weeks and much lower than the 10+ weeks level we typically see ahead of downcycles. Given the limited capacity expansion and supply growth relative to demand growth that we forecast during the next 12–18 months, we expect this low level of inventory to likely continue.

For the customers, despite the aggressive memory procurement during the past few quarters, we believe the inventory level will remain at around a normal level as available supply is used for immediate production, especially for server memory. Given the healthy inventory level across the market, in addition to the increased LTA coverage we are seeing and expecting, we believe an abrupt reversal of the current memory pricing trend would be unlikely.

Exhibit 8: We estimate inventory levels for DRAM to be within 2-4 weeks, below the normal level of around 4-5 weeks

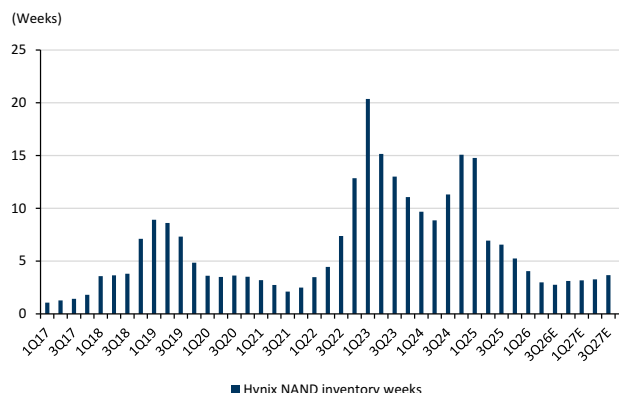
Hynix DRAM inventory level



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 9: We estimate inventory levels for NAND to be within 2-4 weeks, below the normal level of around 4-5 weeks

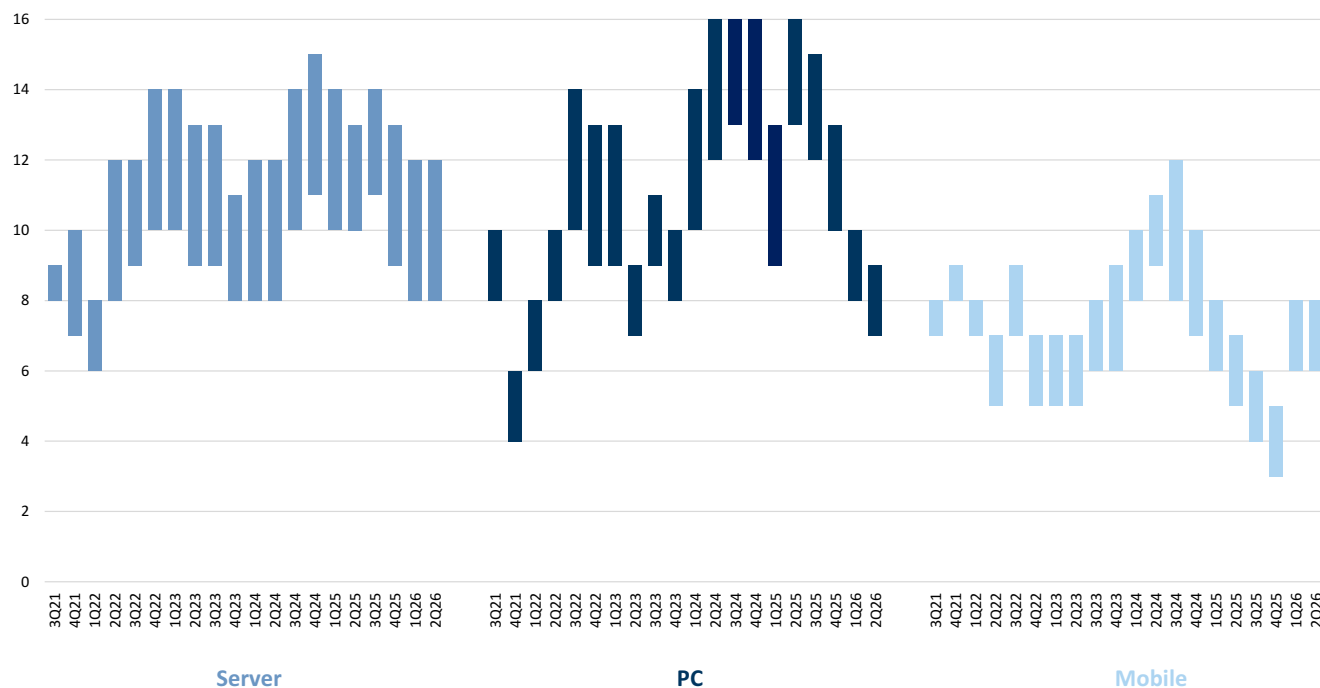
Hynix NAND inventory level



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 10: DRAM customer inventory level

(weeks)



Source: Goldman Sachs Global Investment Research

Q4. Is NAND turning into oversupply?

In addition to the recent concern around inventory at module makers, there have also been some concerns in the market that we are getting closer to an oversupply situation soon that could lead to a pricing decline, especially for NAND. This view is also amplified by the bears pointing to the recent weakness in NAND spot pricing.

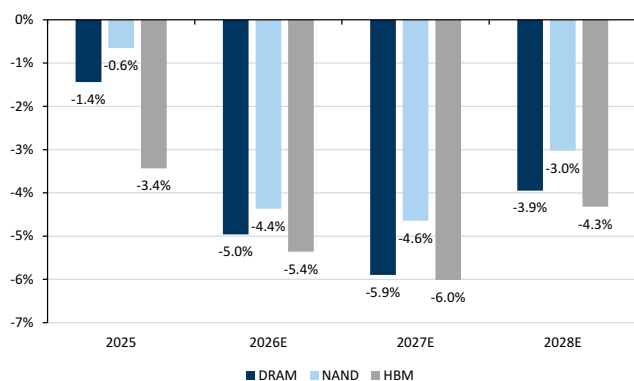
While our view continues to be that DRAM/HBM markets will see slightly tighter S/D compared to NAND especially in 2027, we expect NAND S/D to also see a larger supply tightness in 2027 relative to this year.

We believe the main demand driver for NAND to be AI/server, where we currently estimate 2026E/2027E/2028E enterprise SSD demand of roughly 474EB/619EB/755EB, which implies 66%/31%/22% yoy growth. While recently there has been rising concern over weak consumer demand, based on our checks we continue to sense upside in enterprise SSD demand even during the past few months, which is likely to offset any downside in consumer demand.

In addition, considering our view that major suppliers have focused on DRAM spending over NAND, and also given that we expect the continued focus in NAND to be on tech migration rather than wafer capacity increase, we expect NAND supply to undergrow demand in the medium term.

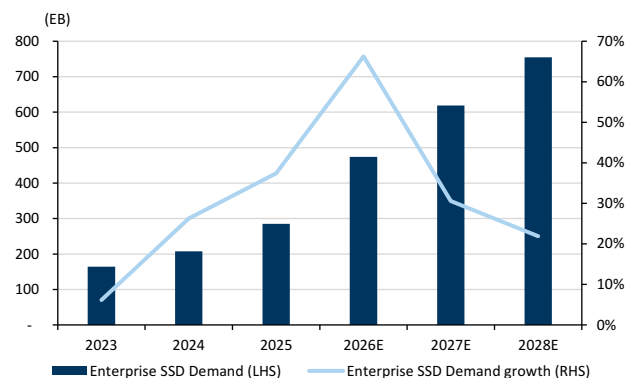
The recent weakness in spot pricing was also centered around a specific product (TLC 512Gb), while other products including TLC 1Tb have been exhibiting a more steady trend. We also note that TLC 512Gb pricing has risen by close to 600% yoy, compared to most other products rising more than 400% yoy. Hence we note that the recent price correction of the TLC 512Gb NAND die is coming after the significant outperformance relative to other products during the past one year.

Exhibit 11: While we expect DRAM/HBM markets will see slightly tighter S/D compared to NAND especially in 2027, we expect NAND S/D to also see a larger supply tightness in 2027 relative to 2026
DRAM, HBM, NAND S/D



Source: Company data, Goldman Sachs Global Investment Research

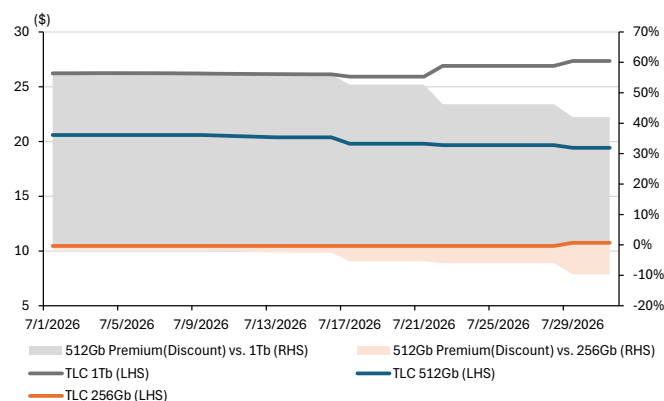
Exhibit 12: We estimate 2026E/2027E/2028E enterprise SSD demand of roughly 474EB/619EB/755EB, which implies 66%/31%/22% yoy growth
Enterprise SSD demand and yoy growth



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 13: Recent weakness in spot pricing was centered around a specific product (TLC 512Gb), while other products including TLC 1Tb have been exhibiting a steady trend

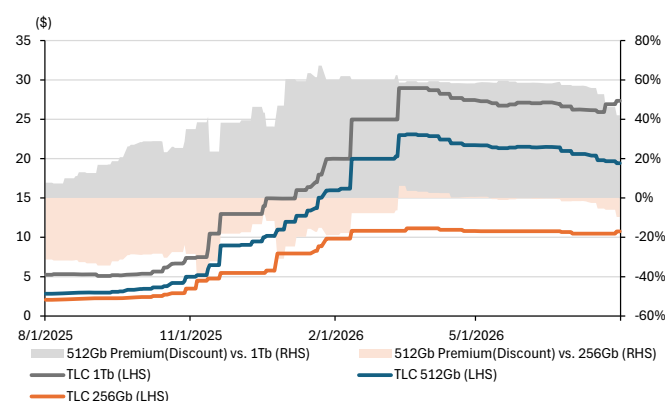
TLC NAND 1 month pricing trend



Source: Bloomberg

Exhibit 14: We note that the recent price correction of the TLC 512Gb NAND die is coming after the significant outperformance relative to other products during the past year

TLC NAND 1 year pricing trend



Source: Bloomberg

Q5. How will CXMT impact the ongoing memory supply shortage?

With CXMT's recent IPO, we are sensing growing investors' concerns around how Chinese memory makers, especially CXMT, will impact the ongoing memory supply shortage. We believe CXMT will increase its presence by primarily focusing on satisfying domestic demand, while export usage is likely to be driven by both commercial and geopolitical considerations. The company could also increase its global presence via Chinese brands' global expansion. However, we think it will not have a material impact on the global supply/demand tightness in the near-to-medium term especially considering the gap in technology level and the geopolitical factors.

CXMT has gained market share in the mobile DRAM space on the back of local China demand, where local smartphone makers have increased procurement from the Chinese DRAM maker due to the tight memory supply. Most of this supply has been coming from LPDDR4(X), where based on TrendForce data the company is expected to ship roughly 70% of its mobile DRAM bits on LPDDR4(X) this year, which compares to global players such as SEC and Hynix having moved to predominantly LPDDR5(X), where exposure in this segment is expected to reach around 75%-85% of their mobile DRAM bits this year, per Trendforce.

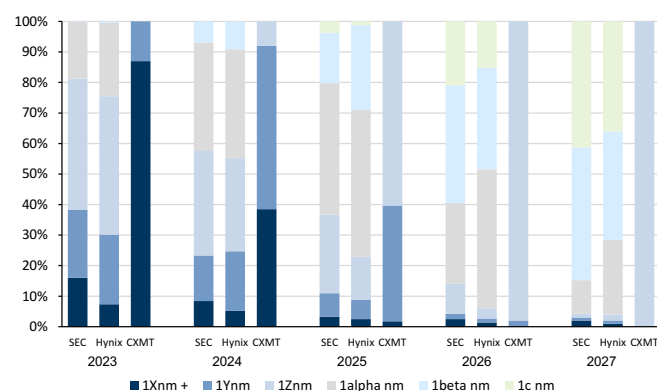
In addition, as seen in the below exhibit, based on TrendForce data CXMT's main technology is roughly equivalent to 1z node, while the global players are in the transition phase from 1a/1b node to 1c node. Hence, there is a technological gap of around 2-3 generations, and considering historically it took around 2-3 years to migrate to the next generation, we believe there is still a significant technological gap between the global players and CXMT.

Going forward, while Korean players will encounter increasing difficulty in tech migration as chip size gets smaller, we believe CXMT will also face similar difficulties as more advanced nodes require advanced WFEs such as EUV, and Chinese semiconductor companies currently have limited access in procuring WFEs including EUV tools due to restrictions. Global DRAM suppliers have started to adopt EUV layers from several years ago, with Hynix mainly starting from 1a nm and SEC even before 1a nm. Hence, while

there's theoretically room for CXMT to advance its technology further, we believe continued restriction of EUV procurement and lengthy periods in developing domestic lithography tools equivalent of EUV performance could act as a bottleneck in catching up to global players over the long term.

Regarding HBM, we believe there is still a meaningful gap compared to the leading players on data transfer speed, power efficiency, and production yield. We believe catching up on HBM technology will be more difficult than conventional DRAM, given the substantial packaging processes added on top of conventional DRAM. Moreover, as HBM generations evolve, advanced foundry processes are required for the base die, hence we believe for CXMT to secure HBM competitiveness, it will also require local Chinese foundry players to improve competitiveness in their advanced nodes.

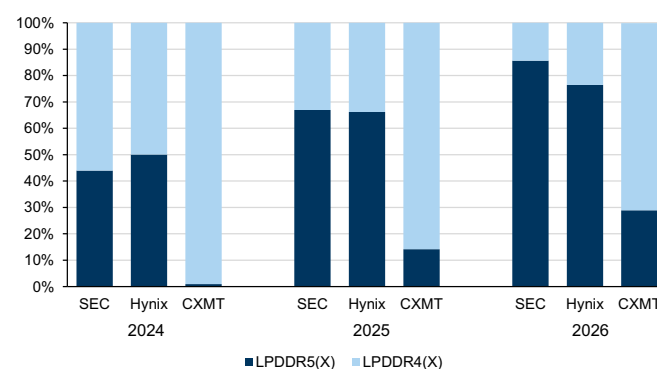
Exhibit 15: We believe there is still a significant technological gap between the global players and CXMT
Portion of DRAM technology node by company



SEC and Hynix based on GSe, CXMT based on TrendForce

Source: Goldman Sachs Global Investment Research, TrendForce

Exhibit 16: We believe there is still a significant technological gap between the global players and CXMT
Portion of DRAM generation by company



SEC and Hynix based on GSe, CXMT based on TrendForce

Source: Goldman Sachs Global Investment Research, TrendForce

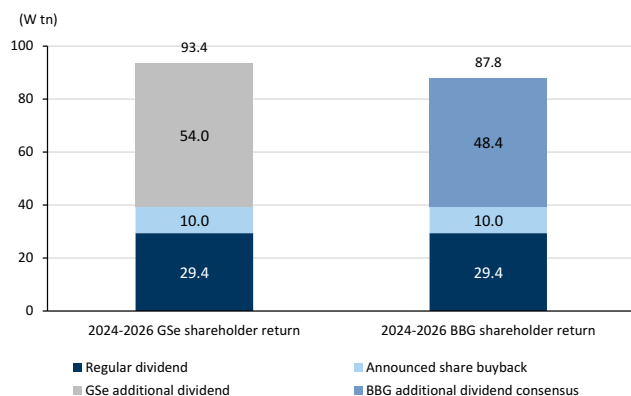
Q6. How will Korean players respond to increasing shareholder return demand?

As Korean players are generating strong cash flow from their core business plus sizable non-operating cash flow for Hynix due to its U.S. ADR listing and Kioxia equity investment, investors' demand for shareholder return has become clear, and we sensed there was some disappointment from the recent earnings calls due to the lack of concrete announcements. Notably, on August 3 which was the first trading day following Kioxia's shareholder return announcement, its share price rose 6% while Korea players share prices saw declines (SEC -9%, Hynix -9%), which we believe at least part of the reason for the divergence was the difference in shareholder return.

Despite the absence of new announcements last week, both SEC and Hynix showed their willingness to deliver promised shareholder returns, acknowledged they are aware of the market's strong interest in shareholder returns, and signaled that they are actively reviewing various options to return to the shareholders. For SEC, of which its 3-yr shareholder return policy expires this year, we believe there needs to be an increase to this year's DPS consensus (W8,638) to fulfill promised shareholder returns (target to return 50% of 3-yr FCF) and see upside to our updated 2026 DPS (W9,500) based on consensus FCF. For Hynix, of which its 3-yr shareholder return policy spans from 2025 to 2027, we also see upside to consensus and our DPS estimates considering the huge

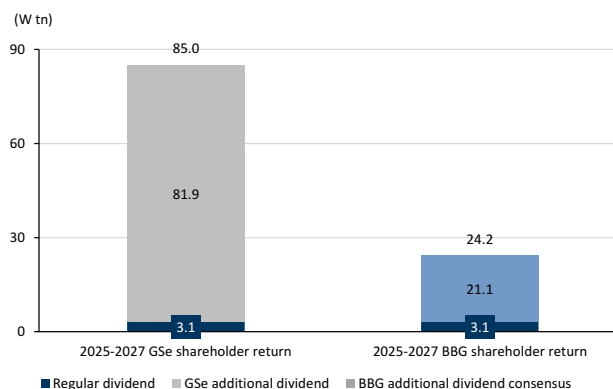
amount of free cash flow we are expecting the company to generate. In addition to additional dividends, we believe potential buyback announcements will be strongly welcomed by the market as the share prices have seen material correction over the last few weeks. For Hynix, given the ADR listing led to share dilution, executing a buyback and cancellation could be one of the options to offset the impact.

Exhibit 17: We believe there needs to be an increase to this year's DPS consensus (W8,638) to fulfill promised shareholder returns
SEC shareholder returns



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 18: We see upside to consensus and our DPS estimates considering the huge amount of free cash flow we are expecting the company to generate
Hynix shareholder returns



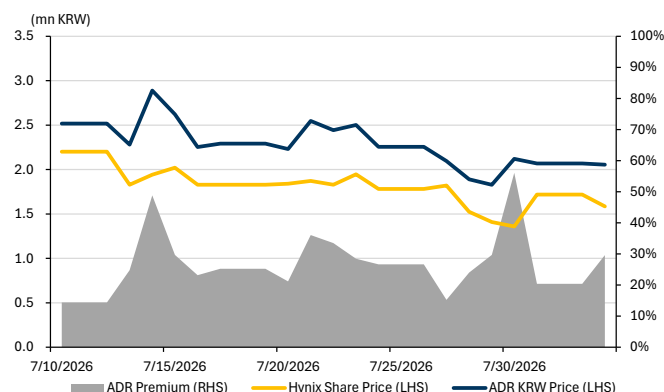
Source: Company data, Goldman Sachs Global Investment Research

Q7. How will Hynix's U.S. ADR impact domestic shares?

Regarding Hynix's ADR, investors' questions center around whether the ADR can help close Hynix's valuation discount versus U.S.-listed memory peers such as Micron and how Hynix's U.S. ADR will trade compared to its domestic shares. Following its ADR listing on 10 July, Hynix ADR has been consistently trading at premium vs. its domestic shares and there continues to be valuation gap with Micron. We believe the main reasons for the discrepancy include: 1) limited fungibility leading to different investor base and 2) limited ADR issuance (2.4% of total shares issued). Hynix said that while the ADRs can be freely converted into domestic shares, conversion of domestic shares into ADRs may be limited by procedures and conversion limits as the issuer may need to go through regulatory filing procedures, which are expected to take several weeks or longer and total ADR issuance cannot exceed the conversion limit of 177.9mn (equivalent of 17.79mn domestic shares) which is the number of ADRs issued.

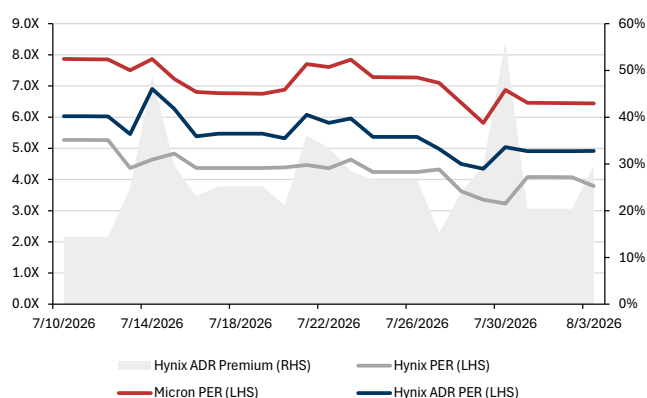
Hynix chairman Mr. Chey Tae-won said the company is open to issuing additional U.S. ADRs. While this could further narrow the gap between U.S. ADR and domestic shares, referring to TSMC's case where the U.S. ADRs represent a higher portion vs. Hynix and has constantly traded at a premium vs. the domestic shares, we expect Hynix's U.S. ADRs will likely trade at a premium over its domestic shares even with additional ADR issuance unless the domestic shares to ADR fungibility limitation is alleviated. At the same time, with the ADRs leading to direct access to global institutional investors, we believe it can help to gradually resolve the historical discount that it has received relative to global peers.

Exhibit 19: Hynix U.S. ADR is trading at a 30% premium vs. domestic shares, with an average premium of 26% since its ADR listing



Source: Bloomberg

Exhibit 20: Hynix domestic shares PER is 41%/30% lower compared to Micron/Hynix ADR
Hynix, Micron, Hynix ADR 12m FWD P/E



Source: Goldman Sachs Global Investment Research, Bloomberg

Q8. Why did Hynix miss 2Q26 and is this trend going to continue in 2H26?

Hynix reported 2Q26 revenue of W79.3tn (+51% qoq, +257% yoy) and operating profit of W60.5tn (+61% qoq, +557% yoy), where OP was broadly inline with GSe of W59.1tn but ~7% lower compared to Bloomberg consensus estimate of W65tn.

We believe the OP miss compared to consensus mainly came from the lower-than-expected DRAM ASP (+29% qoq) during the quarter, as the figure was also lower than our estimate of +39% qoq despite the company's overall OP being inline with our estimate. We believe both conventional DRAM and HBM ASPs were moderately lower than our expectations, with the former likely starting to reflect pricing that were agreed with customers in advance and the latter due to limited mix improvement to HBM4.

We however expect both DRAM bit shipment (+10% qoq) and ASP (+19% qoq) to be solid in 3Q26, as the company starts to ramp HBM4 volume and expand 1c nm DRAM shipment, leading to an improved product mix. This leads to our 3Q26 OP estimate of W77tn, which is broadly inline with consensus estimate. However, depending on the strength in conventional DRAM pricing during the quarter, we believe the company could have higher upside compared to its peers, as we think the company is less exposed to pricing that has been locked-in and includes a ceiling.

Tweak Hynix earnings estimates while maintain TP of W3.5mn and Buy rating

Mainly reflecting 2Q26 results that include a significant investment asset related gain (W63tn), we raise our 2026E EPS estimate by 20%. As we reflect the lower conventional DRAM and NAND pricing estimates that are mostly offset by the substantially higher HBM pricing and slightly higher conventional DRAM and NAND bit shipment estimates, our 2027E–2028E EPS estimates marginally decline by 0–1%. As a result, our 12m TP of W3.5mn is unchanged, which is derived from applying a target P/E of ~9X to the 2026E–2027E average EPS of W416,265. We reiterate our Buy rating on Hynix.

Key risks include 1) major deterioration in memory supply/demand and delay in technology migration, 2) weaker demand for smartphones/PCs/servers which would

impact overall conventional memory demand, 3) Samsung's positive HBM business progress which would impact HBM revenue and profit, 4) lower AI-related capex which would impact overall memory demand, and thus revenue/profit for the company.

Exhibit 21: Hynix DRAM key assumptions

DRAM: Key assumptions	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E
ASP (US\$, 1Gb equiv.)	1.1	1.4	1.7	1.8	2.0	2.1	2.1	2.1	0.2	0.4	0.6	1.5	2.1	2.2
Sequential change (%)	65%	29%	19%	8%	10%	3%	3%	-2%	-37%	77%	28%	172%	36%	3%
Shipment (mn, 1Gb equiv.)	25,430	27,090	29,907	32,599	30,969	32,208	34,784	36,872	64,010	75,110	93,100	115,026	134,833	162,034
Sequential change (%)	0%	7%	10%	9%	-5%	4%	8%	6%	14%	17%	24%	24%	17%	20%
Wafer capacity (WPM, 12-inch equiv.)	580	600	630	660	670	670	680	700	326	400	519	618	680	748
Sequential change (%)	5%	3%	5%	5%	2%	0%	1%	3%	-14%	23%	30%	19%	10%	10%
Sales (W bn)	41,220	58,180	74,322	86,293	85,793	91,901	102,231	106,198	20,849	45,150	74,934	260,015	386,123	473,182
Sequential change (%)	66%	41%	28%	16%	-1%	7%	11%	4%	-27%	117%	66%	247%	49%	23%
Operating profit (W bn)	31,220	46,116	60,341	69,971	70,471	75,060	83,127	85,428	355	20,986	44,953	207,647	314,087	372,321
Sequential change (%)	84%	48%	31%	16%	1%	7%	11%	3%	-96%	5817%	114%	362%	51%	19%
OP Margin (%)	76%	79%	81%	81%	82%	82%	81%	80%	2%	46%	60%	80%	81%	79%

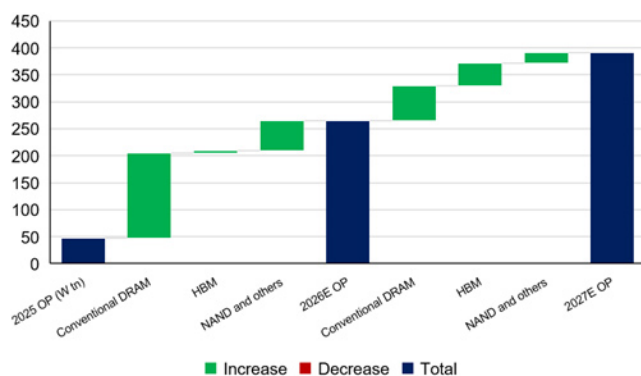
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 22: Hynix NAND key assumptions

NAND: Key assumptions	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E
ASP (US\$, 16Gb equiv.)	0.3	0.5	0.6	0.7	0.7	0.7	0.7	0.6	0.1	0.2	0.2	0.6	0.7	0.6
Sequential change (%)	71%	56%	17%	9%	4%	2%	-6%	-9%	-43%	90%	-8%	242%	22%	-10%
Shipment (mn, 16Gb equiv.)	22,200	25,860	26,636	28,234	26,822	28,968	31,285	32,537	79,500	77,970	87,400	102,930	119,613	141,970
Sequential change (%)	-13%	16%	3%	6%	-5%	8%	8%	4%	18%	-2%	12%	18%	16%	19%
Wafer capacity (WPM, 12-inch equiv.)	240	240	240	250	250	250	260	260	213	205	236	243	255	273
Sequential change (%)	0%	0%	0%	4%	0%	0%	4%	0%	-25%	-4%	15%	3%	5%	7%
Sales (W bn)	11,251	21,004	24,612	28,047	26,363	29,042	29,483	27,903	9,720	18,923	20,409	84,913	112,791	118,382
Sequential change (%)	50%	87%	17%	14%	-6%	10%	2%	-5%	-31%	95%	8%	316%	33%	5%
Operating profit (W bn)	6,421	14,454	17,137	19,945	18,972	20,709	20,194	17,793	(7,808)	2,612	2,608	57,957	77,668	75,266
Sequential change (%)	186%	125%	19%	16%	-5%	9%	-2%	-12%	n.m	n.m	0%	2122%	34%	-3%
OP Margin (%)	57%	69%	70%	71%	72%	71%	68%	64%	-80%	14%	13%	68%	69%	64%

Source: Company data, Goldman Sachs Global Investment Research

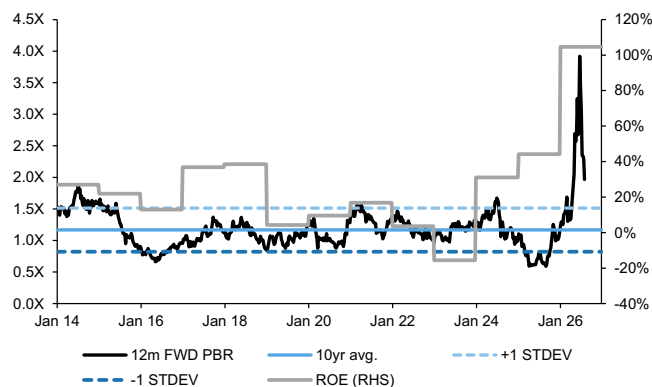
Exhibit 23: Hynix OP waterfall chart



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 24: Hynix shares are trading at 2.0X 12m FWD P/B with 2026E/2027E ROE of 105%/58%

Hynix 12m FWD P/B



Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 25: Hynix earnings revisions

(W bn)	Revised				Previous				% change			
	2Q26	FY26E	FY27E	FY28E	2Q26E	FY26E	FY27E	FY28E	2Q26	FY26E	FY27E	FY28E
Sales	79,319	345,481	499,592	592,400	82,396	357,014	506,881	589,163	-3.7%	-3.2%	-1.4%	0.5%
DRAM	58,180	260,015	386,123	473,182	61,803	269,874	387,061	463,262	-5.9%	-3.7%	-0.2%	2.1%
NAND	21,004	84,913	112,791	118,382	20,440	86,528	119,050	124,952	2.8%	-1.9%	-5.3%	-5.3%
Others	135	552	678	836	153	613	770	949	-12.0%	-9.9%	-12.0%	-12.0%
Operating profit	60,543	265,484	391,722	447,545	59,113	271,061	400,661	453,901	2.4%	-2.1%	-2.2%	-1.4%
DRAM	46,116	207,647	314,087	372,321	46,054	213,968	320,467	373,660	0.1%	-3.0%	-2.0%	-0.4%
NAND	14,454	57,957	77,668	75,266	13,089	57,225	80,232	80,288	10.4%	1.3%	-3.2%	-6.3%
Others	(27)	(120)	(34)	(42)	(31)	(132)	(39)	(47)	n.m.	n.m.	n.m.	n.m.
EBITDA	64,573	281,904	412,442	473,066	63,042	287,177	420,977	479,017	2.4%	-1.8%	-2.0%	-1.2%
OP Margin (%)	76.3%	76.8%	78.4%	75.5%	71.7%	75.9%	79.0%	77.0%				
DRAM	79.3%	79.9%	81.3%	78.7%	74.5%	79.3%	82.8%	80.7%				
NAND	68.8%	68.3%	68.9%	63.6%	64.0%	66.1%	67.4%	64.3%				
Others	-20.0%	-21.7%	-5.0%	-5.0%	-20.0%	-21.5%	-5.0%	-5.0%				
EBITDA	81%	82%	83%	80%	77%	80%	83%	81%				
Pre-tax profit	122,708	350,791	411,622	470,194	62,002	293,956	415,635	472,934	97.9%	19.3%	-1.0%	-0.6%
Net profit	93,820	266,495	308,708	352,638	46,500	222,079	311,719	354,693	101.8%	20.0%	-1.0%	-0.6%
EPS (W), basic	135,792	385,716	446,813	510,395	67,302	321,429	451,170	513,369	101.8%	20.0%	-1.0%	-0.6%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 26: Hynix earnings summary

(W bn)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E
Revenue	52,576	79,319	99,084	114,501	112,314	121,106	131,888	134,283	32,766	66,193	97,147	345,481	499,592	592,400
DRAM	41,220	58,180	74,322	86,293	85,793	91,901	102,231	106,198	20,849	45,150	74,934	260,015	386,123	473,182
NAND	11,251	21,004	24,612	28,047	26,363	29,042	29,483	27,903	9,720	18,923	20,409	84,913	112,791	118,382
Others	105	135	151	161	158	163	174	183	2,197	2,121	1,804	552	678	836
Gross profit	41,679	65,991	84,384	97,899	97,297	104,239	113,864	113,955	(533)	31,828	58,691	289,953	429,355	500,861
SG&A	4,069	5,449	6,936	8,015	7,862	8,477	10,551	10,743	7,197	8,361	11,484	24,469	37,633	53,316
Operating profit	37,610	60,543	77,448	89,884	89,435	95,761	103,313	103,212	(7,730)	23,467	47,206	265,484	391,722	447,545
DRAM	31,220	46,116	60,341	69,971	70,471	75,060	83,127	85,428	355	20,986	44,953	207,647	314,087	372,321
NAND	6,421	14,454	17,137	19,945	18,972	20,709	20,194	17,793	(7,808)	2,612	2,608	57,957	77,668	75,266
Others	(31)	(27)	(30)	(32)	(8)	(8)	(9)	(9)	(277)	(130)	(355)	(120)	(34)	(42)
Pre-tax profit	51,617	122,708	82,037	94,429	94,144	100,710	108,317	108,450	(11,658)	23,885	50,466	350,791	411,622	470,194
Net income	40,330	93,820	61,526	70,819	70,606	75,530	81,236	81,336	(9,112)	19,789	42,919	266,495	308,708	352,638
EPS (W)	57,733	135,792	89,050	102,502	102,193	109,320	117,578	117,722	(13,242)	28,719	62,158	385,716	446,813	510,395
Margins (%)														
Gross profit margin	79	83	85	86	87	86	86	85	(2)	48	60	84	86	85
Operating profit margin	72	76	78	79	80	79	78	77	(24)	35	49	77	78	76
DRAM	76	79	81	81	82	82	81	80	2	46	60	80	81	79
NAND	57	69	70	71	72	71	68	64	(80)	14	13	68	69	64
Others	(29)	(20)	(20)	(20)	(5)	(5)	(5)	(5)	(13)	(6)	(20)	(22)	(5)	(5)
Net profit margin	77	118	62	62	63	62	62	61	(28)	30	44	77	62	60
Sequential growth (%)														
Revenue	60	51	25	16	(2)	8	9	2	(27)	102	47	256	45	19
Gross profit	85	58	28	16	(1)	7	9	0	na	na	84	394	48	17
Operating profit	96	61	28	16	(0)	7	8	(0)	na	na	101	462	48	14
Net income	165	133	(34)	15	(0)	7	8	0	na	na	117	521	16	14

Source: Company data, Goldman Sachs Global Investment Research

Revise Samsung Electronics DPS forecasts

We adjust our 2026 DPS estimates from W11,494 to W9,500 while maintaining our estimates for 2027/2028, and as a result our 2027/2028 EPS estimates are changed marginally from W67,344/W78,970 to W67,378/W79,040 respectively.

Price Target Risks and Methodology - SK Hynix Inc.

Valuation methodology: Our 2026E/27E avg. P/E-based 12m TP is W3,500,000, applying a target P/E multiple of 9.0X.

Key risks: Key risks include 1) major deterioration in memory supply/demand and delay in technology migration, 2) weaker demand for smartphones/PCs/servers which would impact overall conventional memory demand, 3) Samsung's positive HBM business progress which would impact HBM revenue and profit, 4) lower AI-related capex which would impact overall HBM demand, and thus HBM revenue/profit for the company.

Investment Thesis - SK Hynix Inc.

SK Hynix (Hynix) is the leader in the HBM market, the second-largest DRAM supplier,

and one of the leading NAND suppliers globally. We expect one of the strongest memory upcycles to pan out throughout 2026, as the incrementally higher AI spending mainly from the hyperscalers and their intentions to continue to do so is driving our view that memory demand from servers (server DRAM, SOCAMM, HBM, and eSSD) will significantly outpace supply, hence we are Buy rated on the name.

Price Target Risks and Methodology – Samsung Electronics

Valuation methodology: Our 12m 2026-2027E EV/EBITDA-based SOTP target price for the common share is W490,000. Our 12-month target price for the preference share is W360,000, which is based on our target preferred-to-common shares discount of 27%, derived from averaging: 1) the preferred discount of the 2-factor model and 2) the average preference share discount to common shares during the past 1 month. We are Buy rated on both the common and preference shares.

Key downside risks: 1) major deterioration in memory supply/demand, 2) sharp contraction in smartphone margins, and 3) mobile OLED market share loss.

Investment Thesis – Samsung Electronics

Samsung Electronics (SEC) is one of the largest tech companies in the world, with the #1 market share in several products including memory chips, OLED panels, smartphones, and TVs. We expect the strong earnings power led by memory to continue, supported by our expectation of solid memory pricing continuing and also remaining at an elevated level. We expect this to be helped by the long-term agreements (LTA) with customers, with a higher level of binding commitment likely this time compared to the past. The company is starting to show meaningful progress in HBM, and also considering the higher scope for meaningful shareholder returns, we remain constructive on the name and are Buy rated.

Disclosure Appendix

Reg AC

We, Giuni Lee, Taeyong Lee and Daiki Takayama, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

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Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for SK Hynix Inc., Samsung Electronics and Samsung Electronics (Pref) is/are relative to the other companies in its/their coverage universe: Hansol Chemical, LG Display, LG Electronics, LG Innotek Co., SK Hynix Inc., SKC, Samsung Electro-Mechanics, Samsung Electronics, Samsung Electronics (Pref)

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

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Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: Samsung Electronics (W240,000), Samsung Electronics (Pref) (W176,400) and SK Hynix Inc. (W1,577,000)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: Samsung Electronics (W240,000), Samsung Electronics (Pref) (W176,400) and SK Hynix Inc. (W1,577,000)

Goldman Sachs had a non-investment banking securities-related services client relationship during the past 12 months with: Samsung Electronics (W240,000) and Samsung Electronics (Pref) (W176,400)

Goldman Sachs had a non-securities services client relationship during the past 12 months with: Samsung Electronics (W240,000), Samsung Electronics (Pref) (W176,400) and SK Hynix Inc. (W1,577,000)

Goldman Sachs has managed or co-managed a public or Rule 144A offering in the past 12 months: SK Hynix Inc. (W1,577,000)

Goldman Sachs makes a market in the securities or derivatives thereof: SK Hynix Inc. (W1,577,000)

Distribution of ratings/investment banking relationships

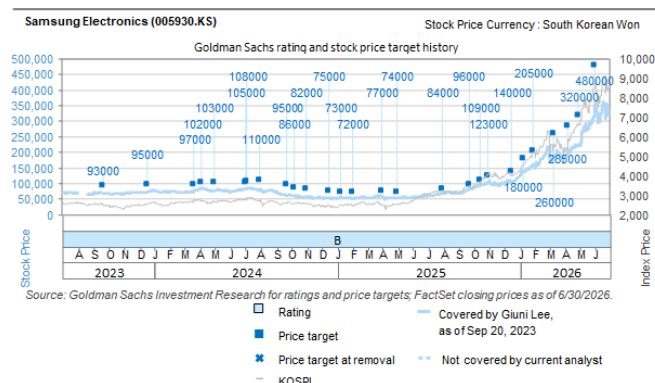
Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution		
	Buy	Hold	Sell
Global	50%	34%	16%

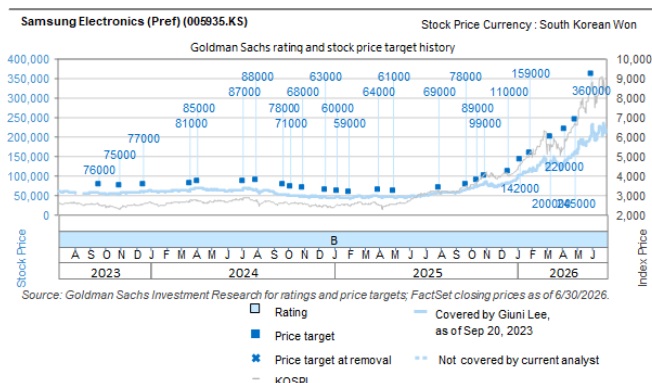
	Investment Banking Relationships		
	Buy	Hold	Sell
	65%	59%	43%

As of July 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,104 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

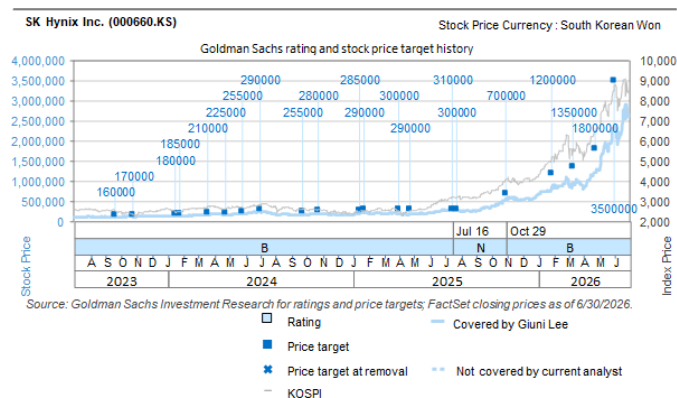
Price target and rating history chart(s)



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



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Target price history table(s)

Samsung Electronics (Pref) (005935.KS)

Date of report	Target price (W)	Closing price (W)
31-May-26	360,000	202,500
30-Apr-26	245,000	158,300
07-Apr-26	220,000	130,900
11-Mar-26	200,000	138,900
29-Jan-26	159,000	115,600
08-Jan-26	142,000	101,900
16-Dec-25	110,000	79,800
30-Oct-25	99,000	82,500
14-Oct-25	89,000	72,300
22-Sep-25	78,000	66,700
31-Jul-25	69,000	57,600
01-May-25	61,000	46,850
01-Apr-25	64,000	47,700

Samsung Electronics (005930.KS)

Date of report	Target price (W)	Closing price (W)
30-Jul-26	490,000	207,000
31-May-26	480,000	317,000
30-Apr-26	320,000	220,500
07-Apr-26	285,000	196,500
11-Mar-26	260,000	190,000
29-Jan-26	205,000	160,700
08-Jan-26	180,000	138,800
16-Dec-25	140,000	102,800
30-Oct-25	123,000	104,100
14-Oct-25	109,000	91,600
22-Sep-25	96,000	83,500
31-Jul-25	84,000	71,400
01-May-25	74,000	55,500

Samsung Electronics (Pref) (005935.KS)

Date of report	Target price (W)	Closing price (W)
02-Feb-25	59,000	43,000
08-Jan-25	60,000	46,800
16-Dec-24	63,000	46,550
31-Oct-24	68,000	47,950
08-Oct-24	71,000	49,900
23-Sep-24	78,000	52,400
31-Jul-24	88,000	64,900
05-Jul-24	87,000	68,000
05-Apr-24	85,000	69,000
21-Mar-24	81,000	65,800
17-Dec-23	77,000	59,300
31-Oct-23	75,000	53,600
20-Sep-23	76,000	56,100

Samsung Electronics (005930.KS)

Date of report	Target price (W)	Closing price (W)
01-Apr-25	77,000	58,800
02-Feb-25	72,000	52,400
08-Jan-25	73,000	57,300
16-Dec-24	75,000	55,600
31-Oct-24	82,000	59,200
08-Oct-24	86,000	60,300
23-Sep-24	95,000	62,600
31-Jul-24	110,000	83,900
05-Jul-24	108,000	87,100
01-Jul-24	105,000	81,800
30-Apr-24	103,000	77,500
05-Apr-24	102,000	84,500
21-Mar-24	97,000	79,300
17-Dec-23	95,000	73,300
20-Sep-23	93,000	69,600

SK Hynix Inc. (000660.KS)

Date of report	Target price (W)	Closing price (W)
31-May-26	3,500,000	2,333,000
23-Apr-26	1,800,000	1,225,000
11-Mar-26	1,350,000	955,000
28-Jan-26	1,200,000	841,000
29-Oct-25	700,000	558,000
24-Jul-25	300,000	269,500
16-Jul-25	310,000	296,000
24-Apr-25	290,000	178,300
01-Apr-25	300,000	197,000
23-Jan-25	290,000	219,500
15-Jan-25	285,000	198,200
24-Oct-24	280,000	198,200
23-Sep-24	255,000	162,000
01-Jul-24	290,000	235,500
27-May-24	255,000	201,500
25-Apr-24	225,000	170,600
21-Mar-24	210,000	170,000
25-Jan-24	185,000	137,400
17-Jan-24	180,000	131,000
26-Oct-23	170,000	120,000
20-Sep-23	160,000	118,000

Price targets shown in table(s) are unadjusted for corporate actions.

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